

Supersonic PINN vs validated modal CSV

Run: assets/pinn_supersonic/singlecase_pq_firstorder_M150_a01625_VALIDATED_anchor_RIGHTTAIL_balancedtail_ft

alpha: 0.1625

Mach: 1.5

p_rel: 0.0167547

q_rel: 0.0246252

p_y_num_rel: 0.0401267

q_vs_dpred_rel: 0.0292055

align_scale_abs: 0.443728

pred_max_abs_p_raw: 1.0022

Zone errors:

center_|y|<=50: p_rel=0.005626, q_rel=0.009675, max|p_pred|=0.4447

center_|y|<=60: p_rel=0.01298, q_rel=0.01146, max|p_pred|=0.4447

diagnostic_|y|<=80: p_rel=0.07613, q_rel=0.06321, max|p_pred|=0.4447

left_-120_-80: p_rel=0.3126, q_rel=0.6459, max|p_pred|=0.2038

right_50_120: p_rel=3.177, q_rel=10.28, max|p_pred|=0.09375

right_y>80: p_rel=20.52, q_rel=58.63, max|p_pred|=0.03175

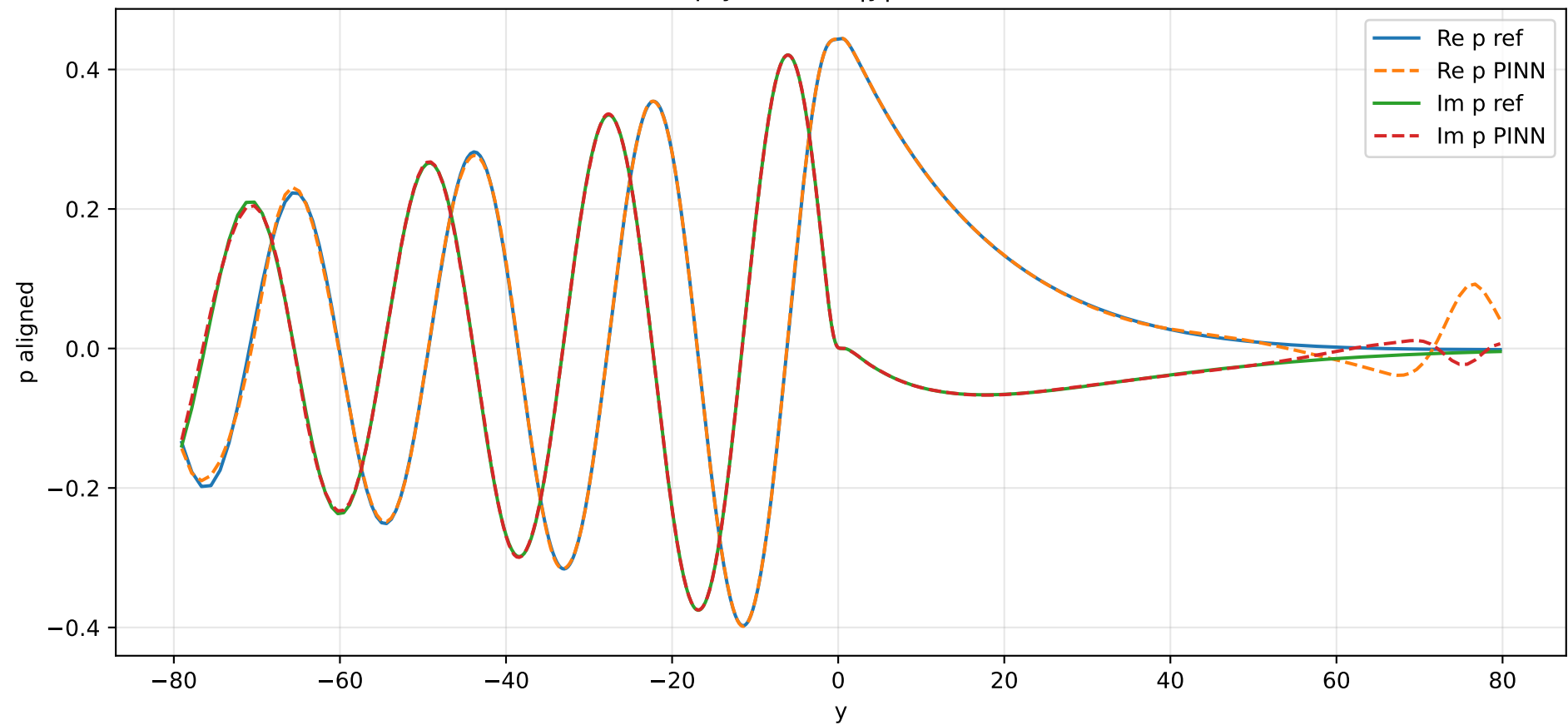
right_y>120: p_rel=203.6, q_rel=569.7, max|p_pred|=0.03175

left_y<-80: p_rel=0.5583, q_rel=0.7049, max|p_pred|=0.2038

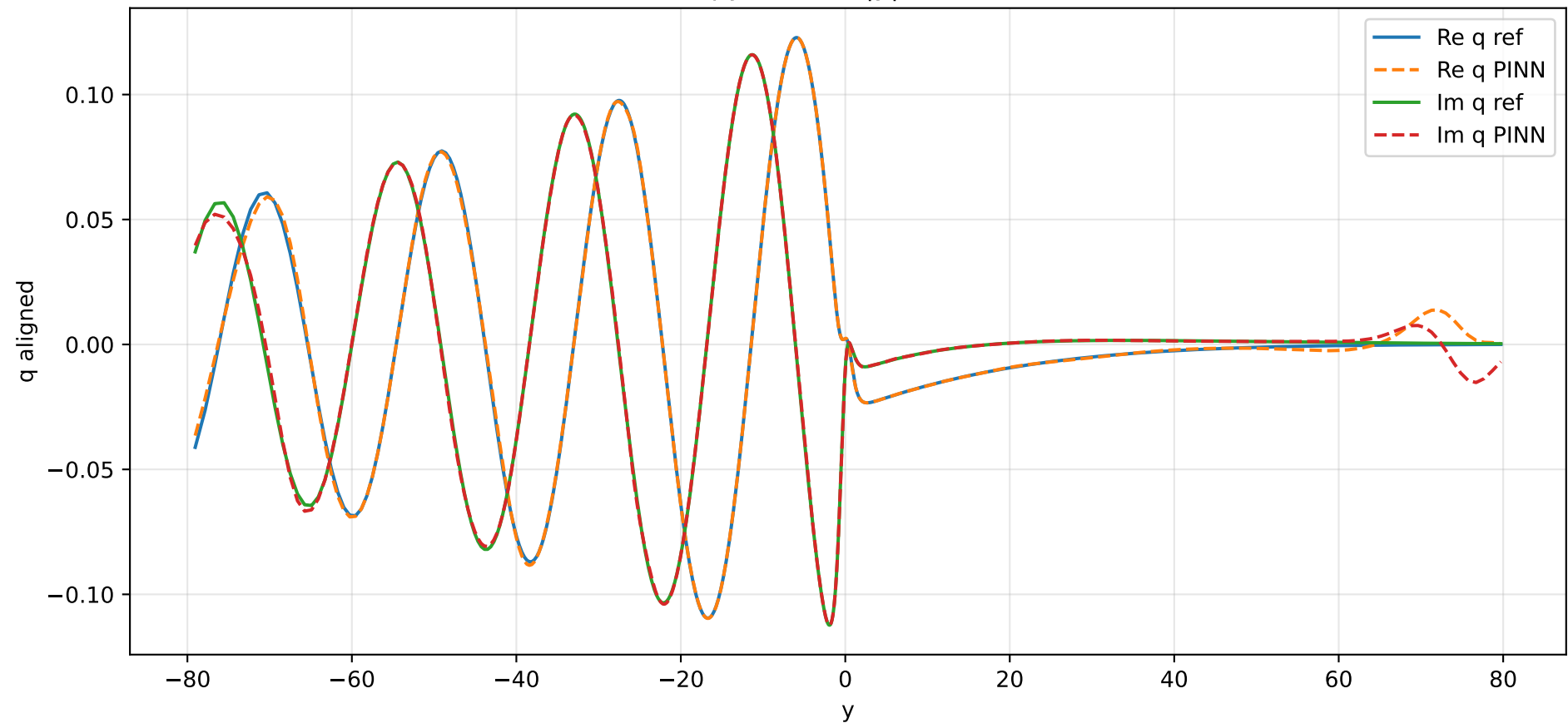
outside_|y|>80: p_rel=0.3732, q_rel=0.4083, max|p_pred|=0.2038

full_domain: p_rel=0.2415, q_rel=0.2761, max|p_pred|=0.4447

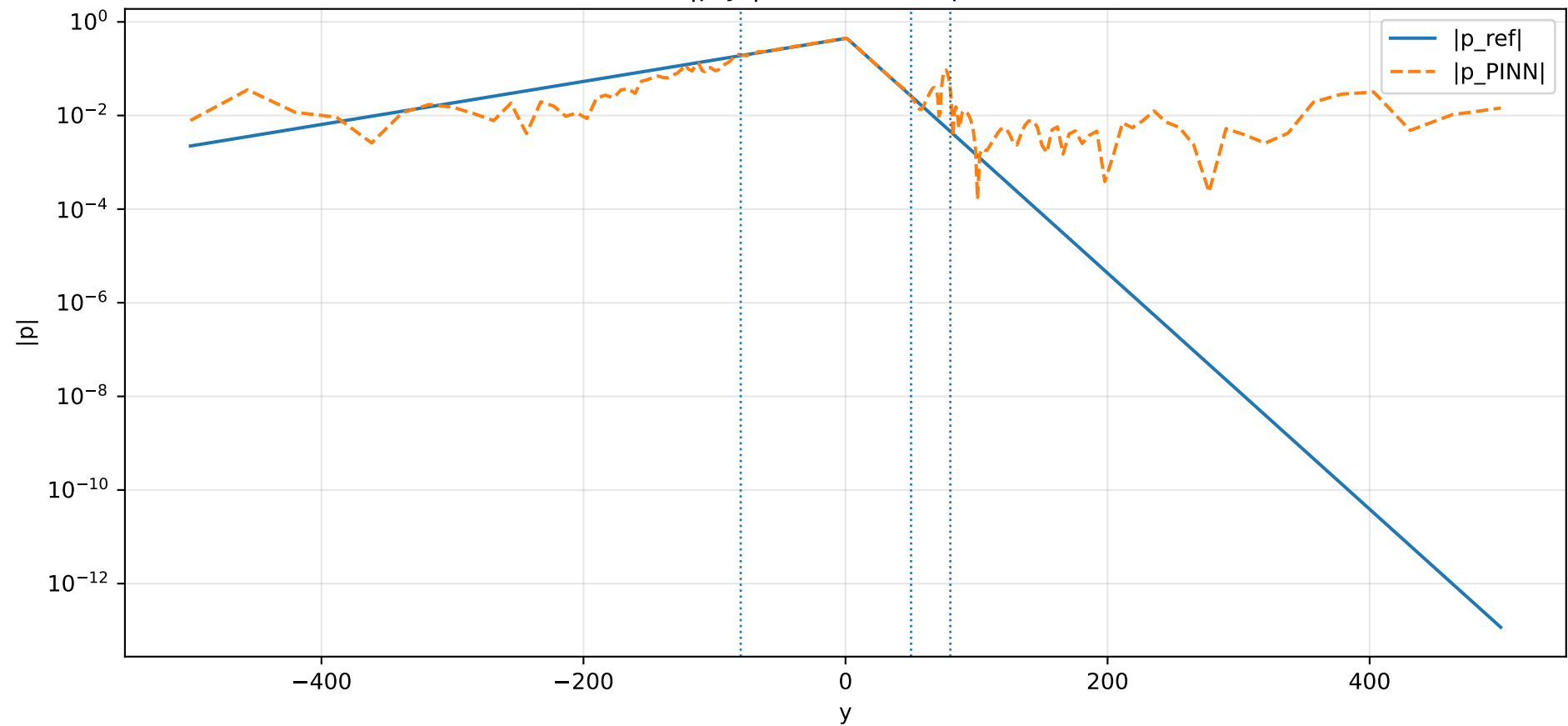
$p(y)$, centre $|y| \leq 80$



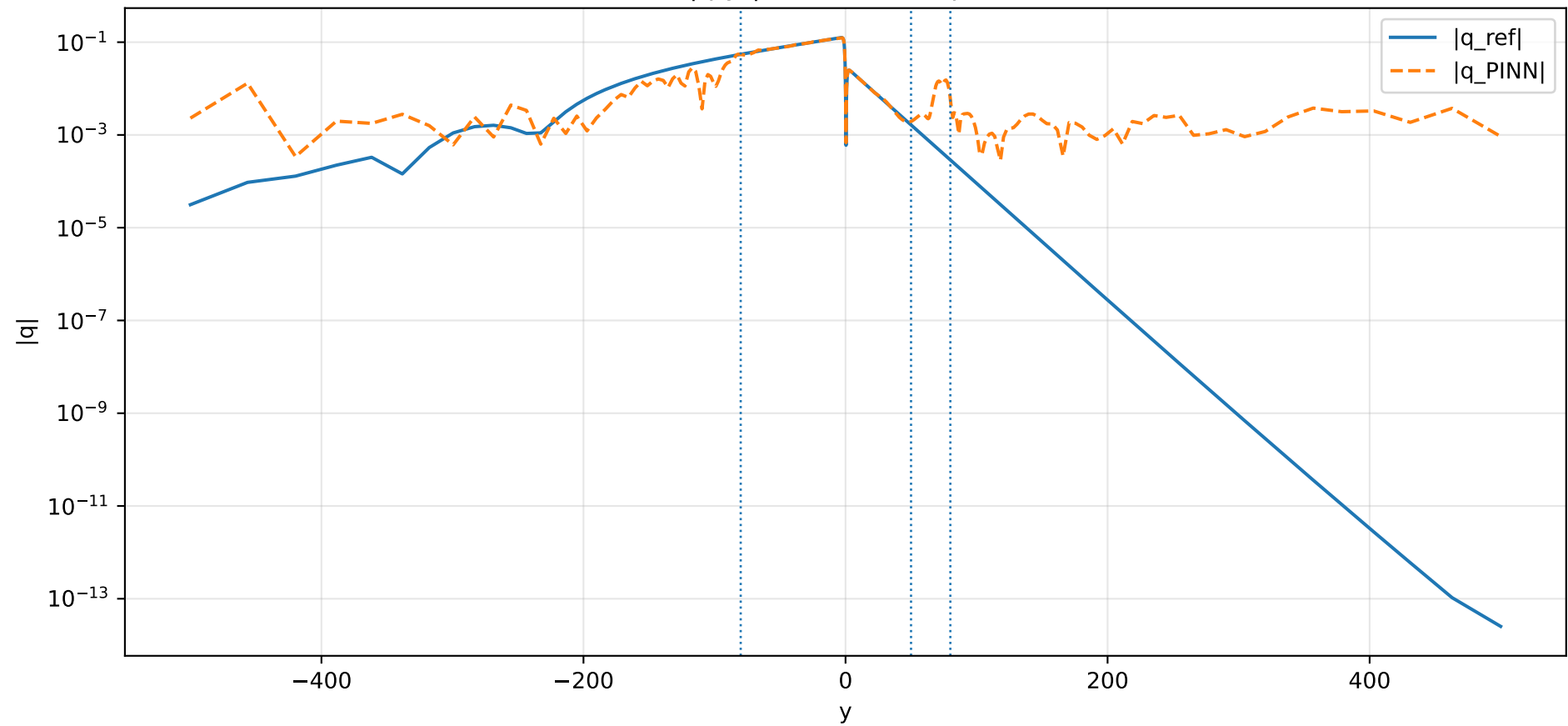
$q(y)$, centre $|y| \leq 80$



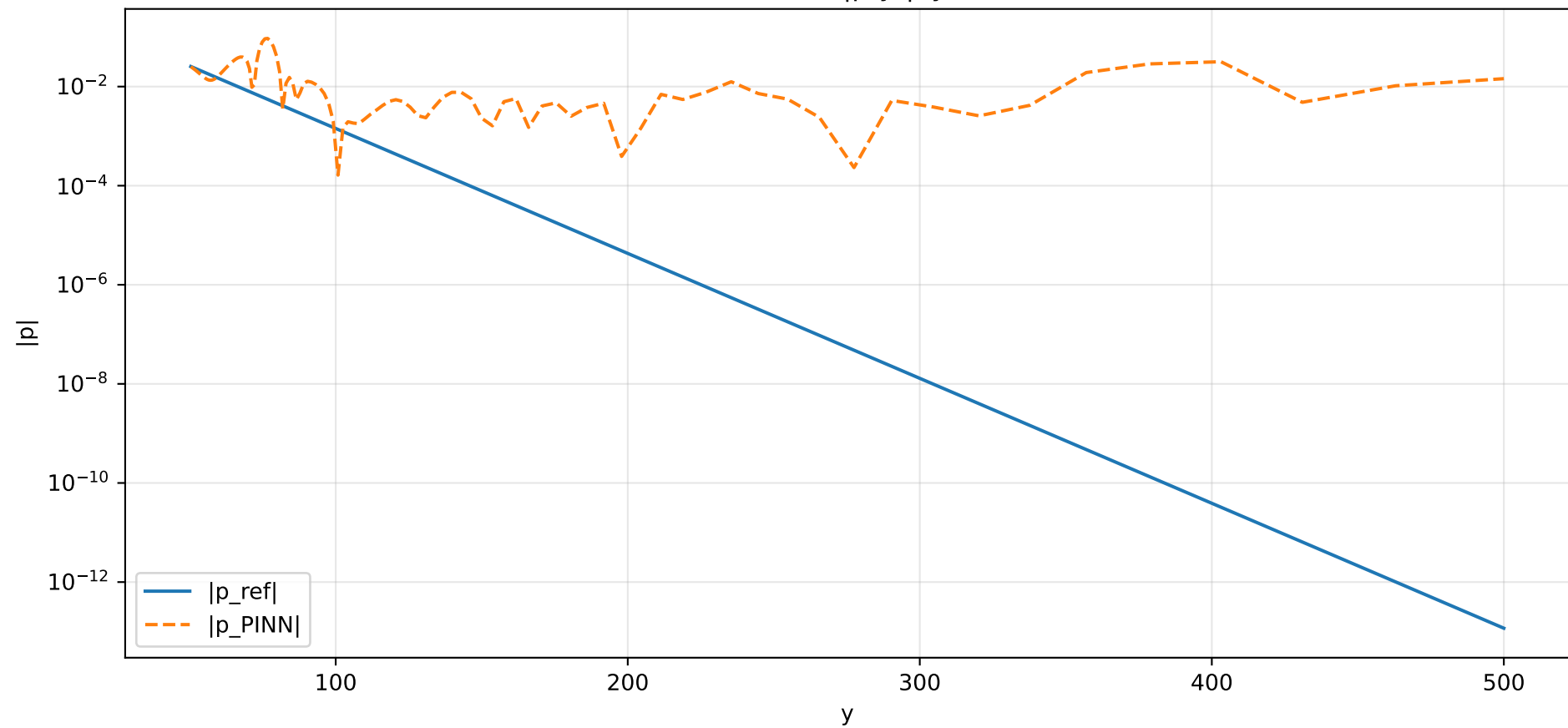
$|p(y)|$ domaine complet



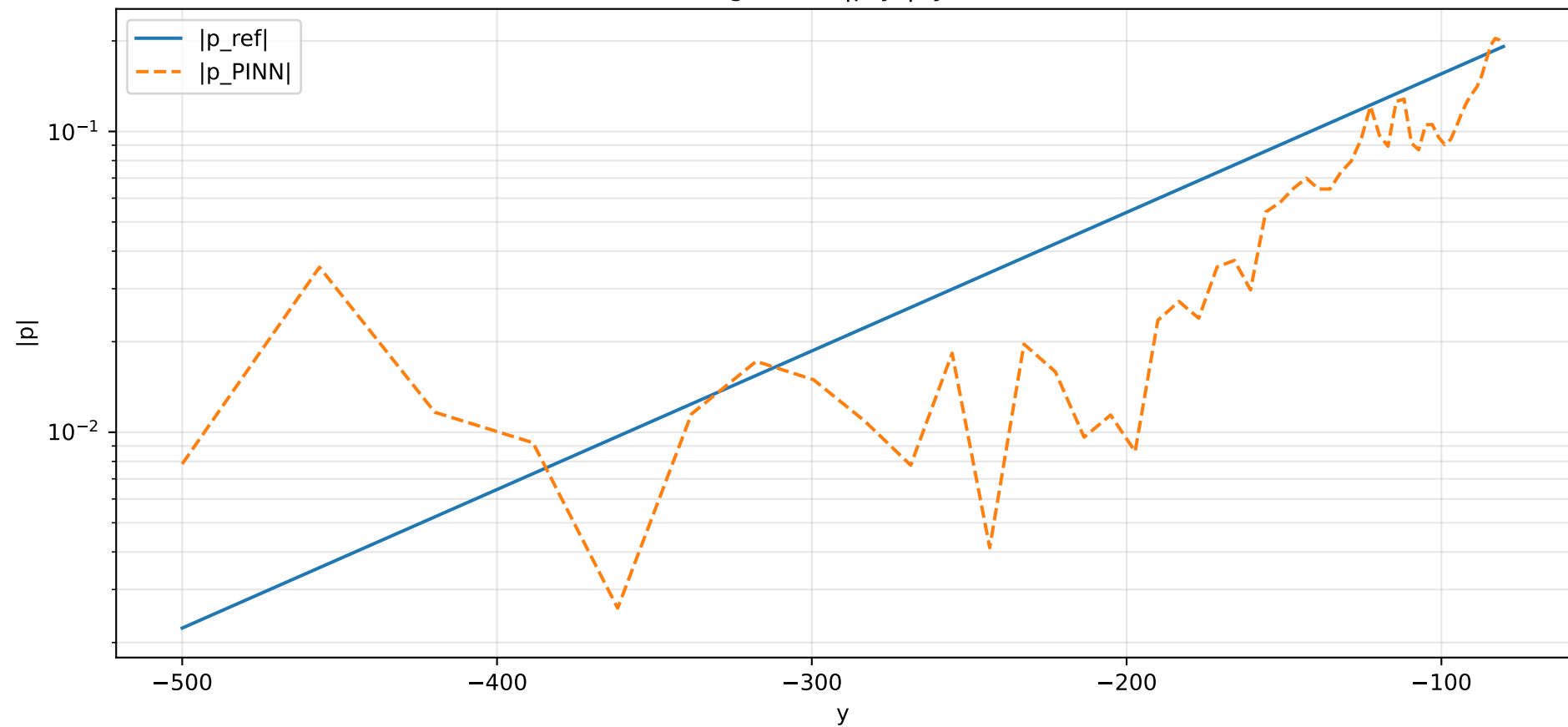
$|q(y)|$ domaine complet



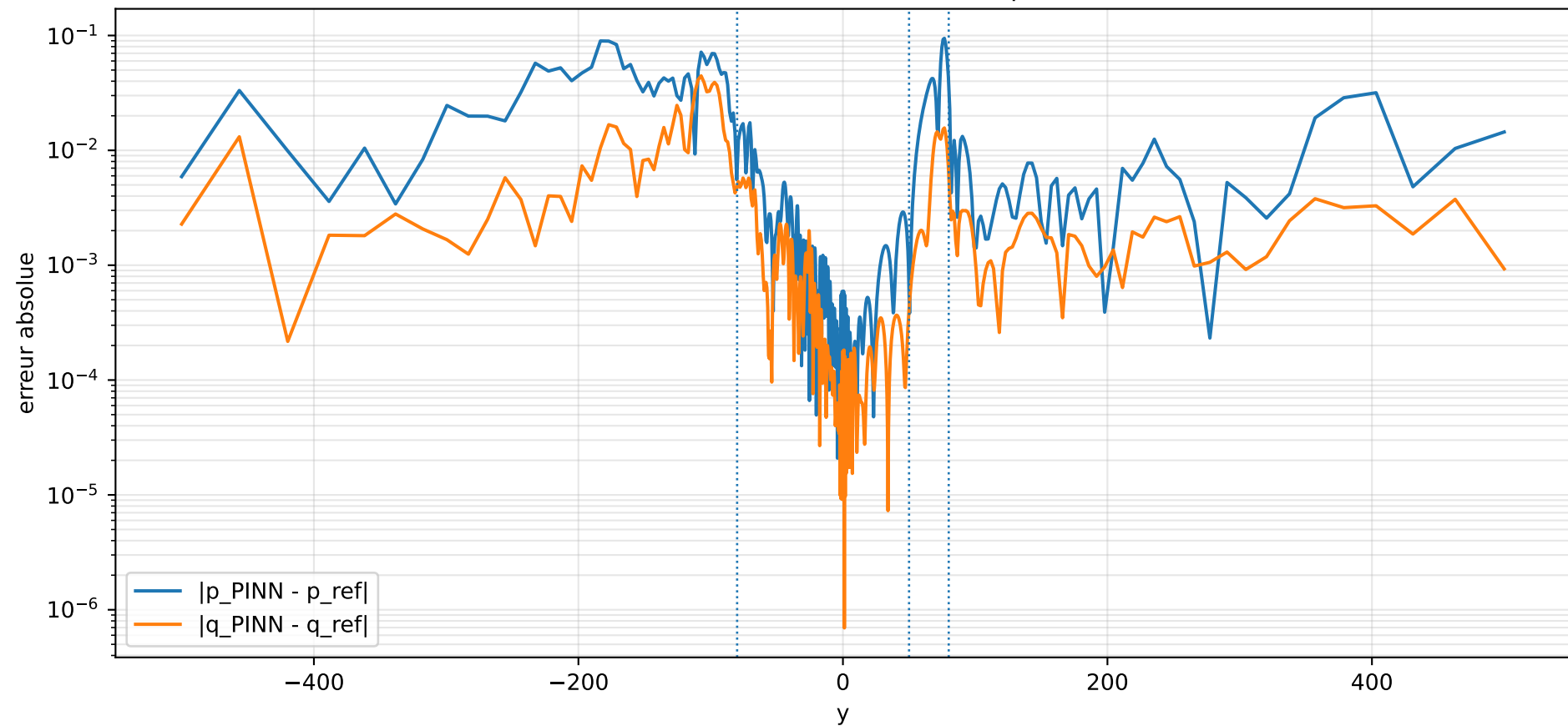
Queue droite, $|p(y)|$, $y \geq 50$



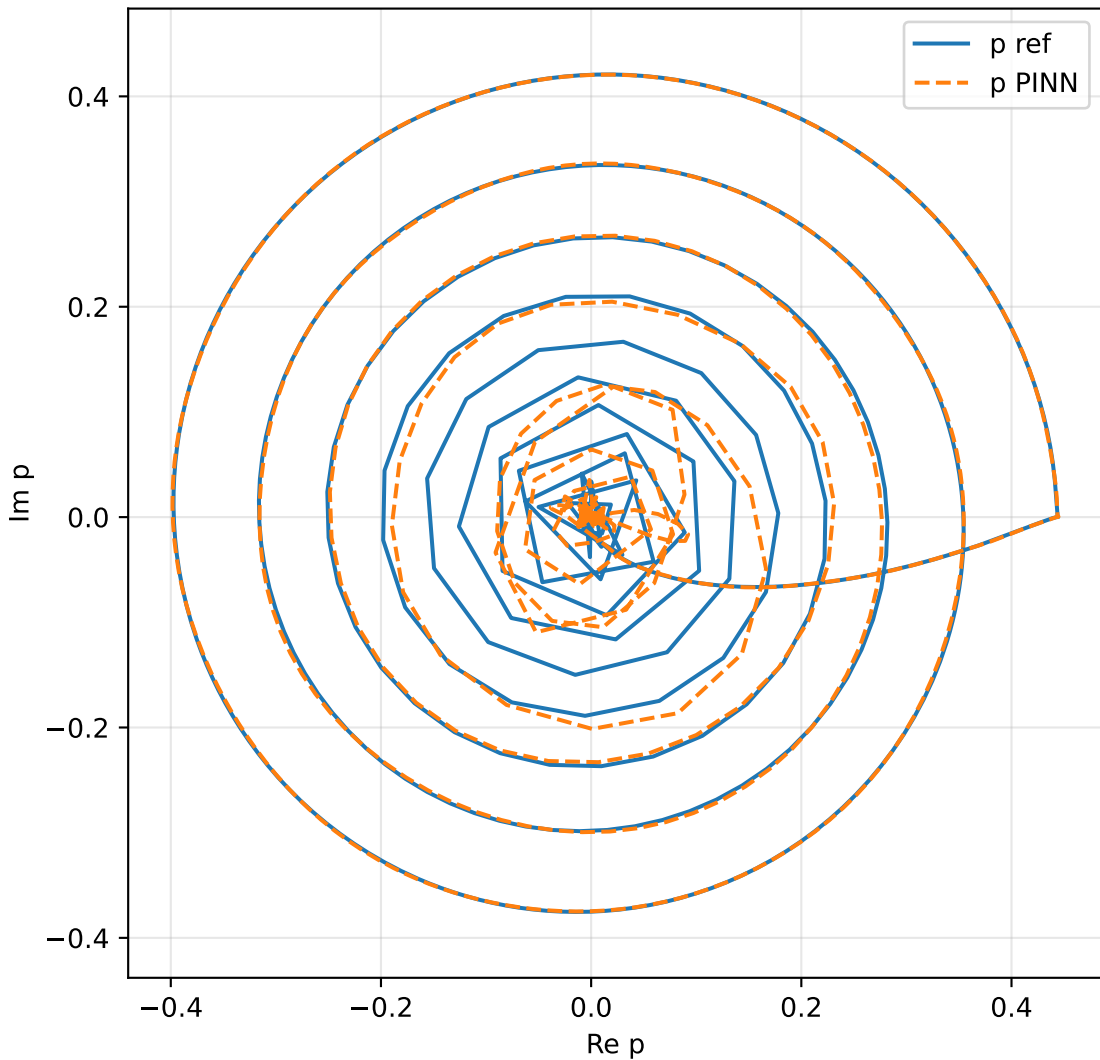
Queue gauche, $|p(y)|$, $y \leq -80$



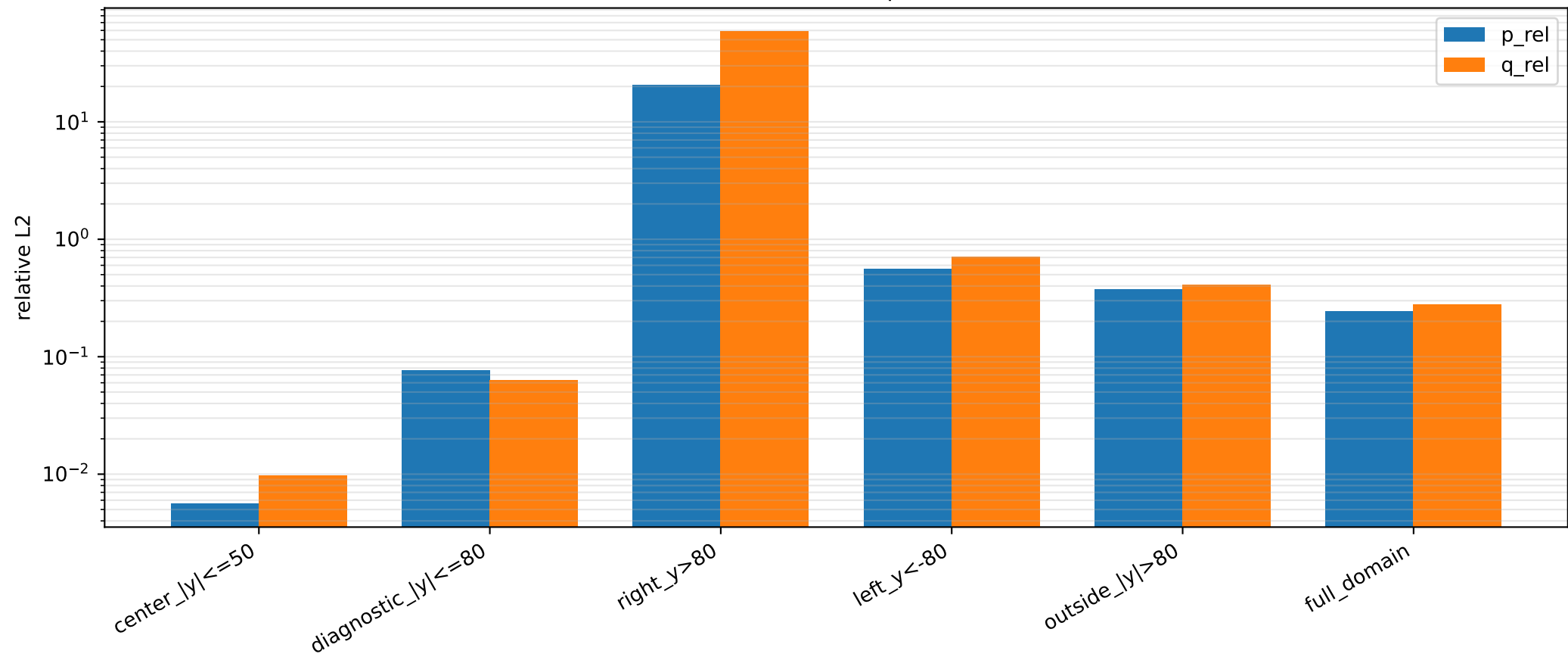
Erreur absolue domaine complet



Trajectoire complexe $p(y)$



Erreurs relatives par zone



Compatibilité $q \approx p_y$, centre $|y| \leq 80$

